Geometrical optics

→ Ray optics simulator: https://phydemo.app/ray-optics/simulator/

Objectives

→ **learn ray optics principles using a simulator to understand the building blocks of fluorescence/confocal microscopy.**

Part 1. Foundations: Light and Lenses

Part 2. Diverging Optics and Imaging

Part 3. Numerical aperture and optical assemblies

Part 4. Imperfections of Lenses

Part 5. Building Toward Microscopy

Report

Concept: Define the key idea (e.g., focal length, numerical aperture, chromatic aberration).

Simulation setup: Describe what you built in the simulator (source, lens type, orientation, detector).

Illustration: Include a screenshot (annotated with arrows/labels) of the optical layout and the resulting rays/intensity.

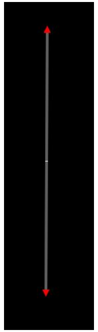
Observation: Explain what you saw .

Figure legend: Provide a clear caption under each screenshot, as if it were a figure in a scientific paper.

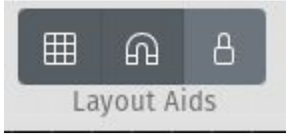
Bonus:

- Point out where ray (geometrical) optics is a simplification and when wave optics (diffraction, interference) becomes essential

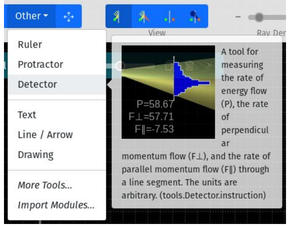
Practical tips



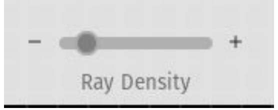
- Start using ideal lenses
- Click on GRID
- Click on Snap To Grid



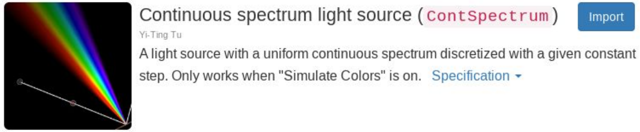
- You can insert a detector anywhere and tick its “Irradiance Map” to see the intensity profile



- Tune “ray density” -/+



- To play with multiple wavelengths:
  - In “Other” ⟶ import modules ⟶ ContSpectrum
  - In Settings ⟶ Simulate colors



Part 1. Foundations: Light and Lenses

1. **Light sources**
   - *Simulation*: Laser (collimated), LED (diverging), fluorophore (isotropic).
   - *Goal*: Compare divergence and isotropy. Relate to excitation vs. emission light in fluorescence microscopy.
2. **Single convex lens with collimated light**
   - *Simulation*: Collimated laser through convex lens.
   - *Goal*: Define **focal length**, **optical axis**, and the principle of focusing.
3. **Single convex lens with divergent light**
   - *Simulation*: LED/fluorophore through convex lens.
   - *Goal*: Show how focal point depends on source position relative to lens.

Part 2. Diverging Optics and Imaging

1. **Concave lens with collimated beam**

   - *Simulation*: Collimated beam diverges through concave lens.

   - *Goal*: Introduce the concept of negative focal length.

2. **Image formation with lenses**

   - *Simulation*: Show how an extended object emits a bundle of rays instead of one.
   - *Simulation*: Object at different distances through convex lens.

   - *Goal*: Bridge the intuition from “one light ray” to “all rays forming an image.” + Compare sharp vs. blurred images; real vs. virtual images.

Part 3. Numerical aperture and optical assemblies

1. **Numerical Aperture (NA)**
    - *Simulation*: Vary aperture size for a convex lens with a point source.
    - *Goal*: Link NA to light collection efficiency.
2. **Beam expanders (telescope configuration)**
    - *Simulation*: Two convex lenses with different focal lengths.
    - *Goal*: Explain beam shaping, collimation, and scaling of beam diameter.

Part 4. Imperfections of Lenses

1. **Plano-convex lens orientation**

   - *Simulation*: Flip plano-convex lens relative to incoming collimated beam.
   - *Goal*: Show practical optical design considerations to minimize spherical aberrations

2. **Chromatic aberrations**

   - *Simulation*: Polychromatic (multi-wavelength) light through a convex lens.
   - *Goal*: Show wavelength-dependent focal lengths; link to need for achromats/apochromats.

Part 5. Building Toward Microscopy

1. **Epifluorescence setup (basic)**
    - *Simulation*: Collimated excitation light focused onto a sample, isotropic fluorescence emission collected by the same lens.
    - *Goal*: Show principle of illumination and collection through a shared objective.

12. **Confocal setup (conceptual)**
    - *Simulation*: Add pinhole or spatial filter at image plane; compare with and without.
    - *Goal*: Show rejection of out-of-focus light; explain optical sectioning.